Experiment 2: Bayes Decision Theory

Title

Spam Message Classification using Bayes Decision Theory

Objective

To classify SMS messages as Spam or Ham using Bayes Theorem and the Naive Bayes algorithm.

Dataset Description

The SMS Spam Collection Dataset contains text messages labeled as:

- Spam
- Ham (Not Spam)

Each record consists of a class label and the SMS message.

Step 1: Load the Dataset

The dataset was loaded using Pandas. The label and message columns were selected for processing.

Step 2: Convert Text into Features

The CountVectorizer converted each SMS message into a numerical feature vector based on word frequencies.

Step 3: Train the Naive Bayes Model

A Multinomial Naive Bayes classifier was trained using the feature vectors and class labels.

Step 4: Test a New Message

A sample message:

"Congratulations! You won a free lottery. Claim now."

was given as input to the trained model.

Step 5: Calculate Posterior Probability

The classifier calculated the posterior probability of the message belonging to both Spam and Ham classes using Bayes Theorem.

The probability of the Spam class was higher than the Ham class.

Analysis

The words "Congratulations", "Free", "Won", "Lottery", and "Claim" frequently appear in spam messages. Therefore, the classifier assigned a higher posterior probability to the Spam class.

Conclusion

The Naive Bayes classifier successfully classified the SMS message as Spam based on Bayes Decision Theory.